Spatial data and data management

Lecture 2 content:

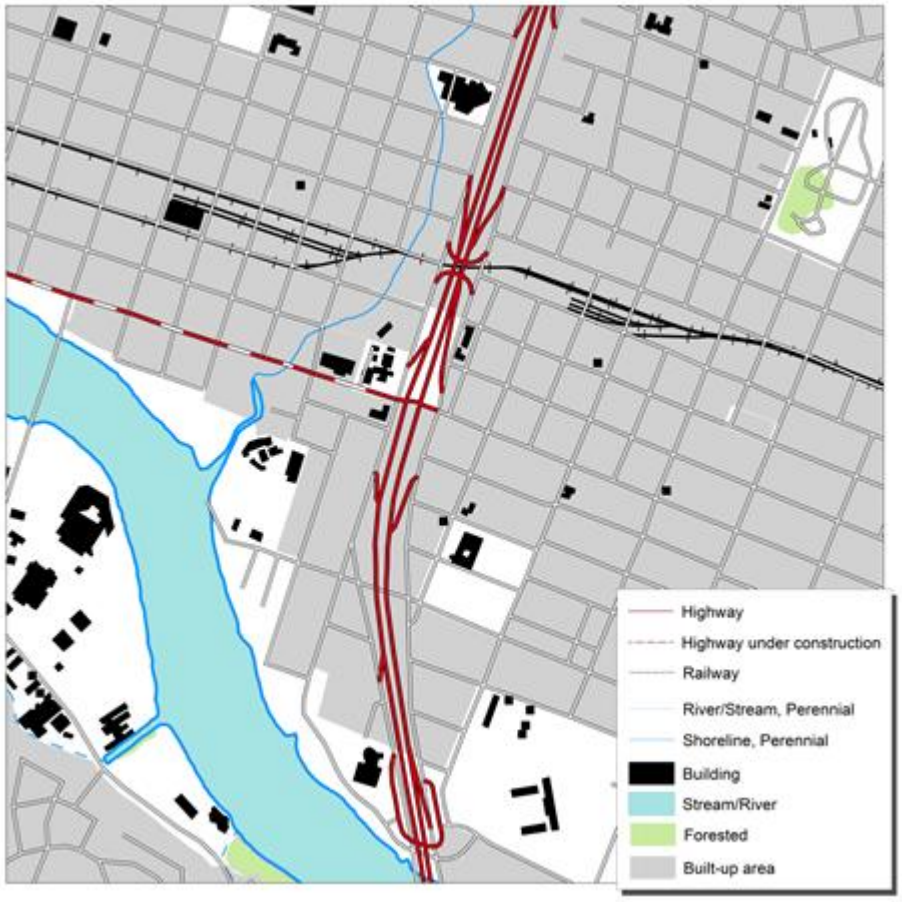
- the characteristics of spatial data
- object and field views and surface models
- understand map scale and cartographic abstraction
- map projections and projected coordinate systems
- how spatial data are represented, structured and organised using computers
- understand differences between vector and raster representations

Nature of spatial data

- Spatial data are usually ‘observational’ rather than ‘experimental’.
- Spatial data capture the complexity of the real world in finite form through a particular process of conceptualisation and representation.
- Spatial data record locations according to a particular georeferencing system.

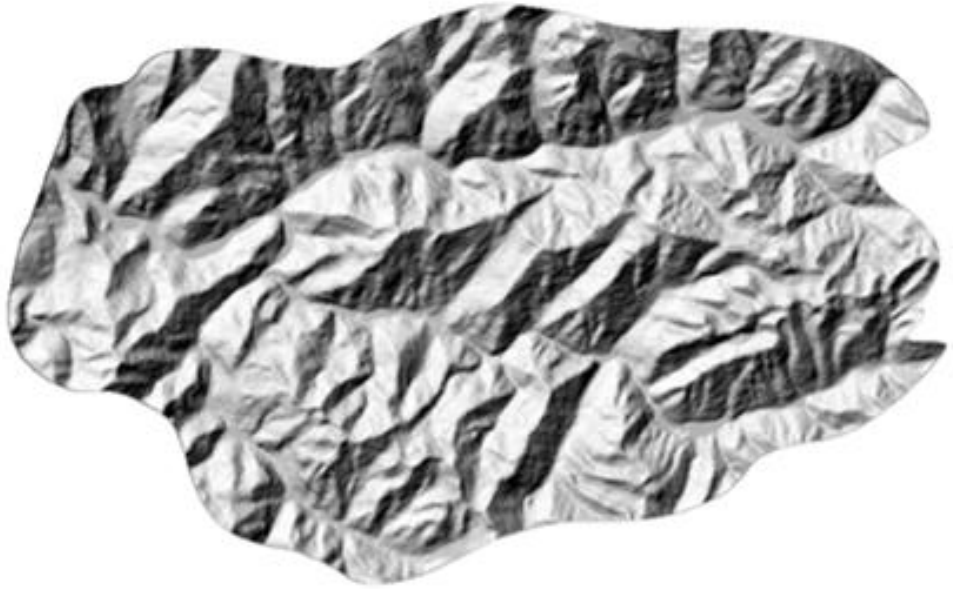
Conceptualisation – Object view

- The object view
  - populated by individual objects.
  - point, line and area objects, or features.
  - exactly located, well-defined boundaries and countable.



Conceptualisation – Field view

- The field view
  - covered by continuous surfaces in which the attribute values can be defined at every location



Surface Models: DEM and DTM

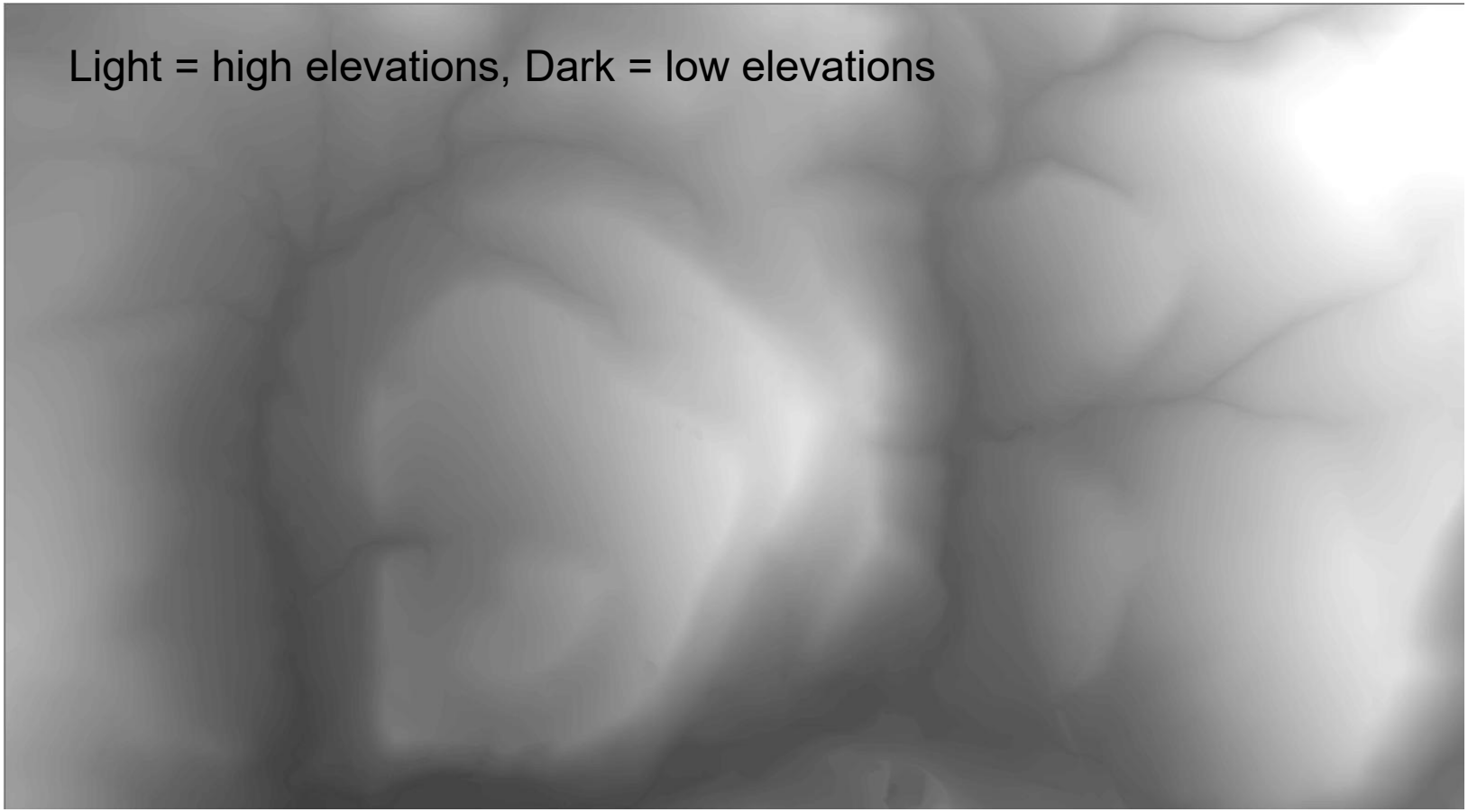
- **Digital Elevation Model (DEM)**
  - A 3D representation of a terrain surface.
  - 'bare-earth' elevation model, meaning it is free of vegetation, man-made structures and other non-terrain objects.
  - DEMs are usually either raster (grid of squares), or vector-based triangular irregular network (TIN) data.

- **Digital Terrain Model (DTM)**

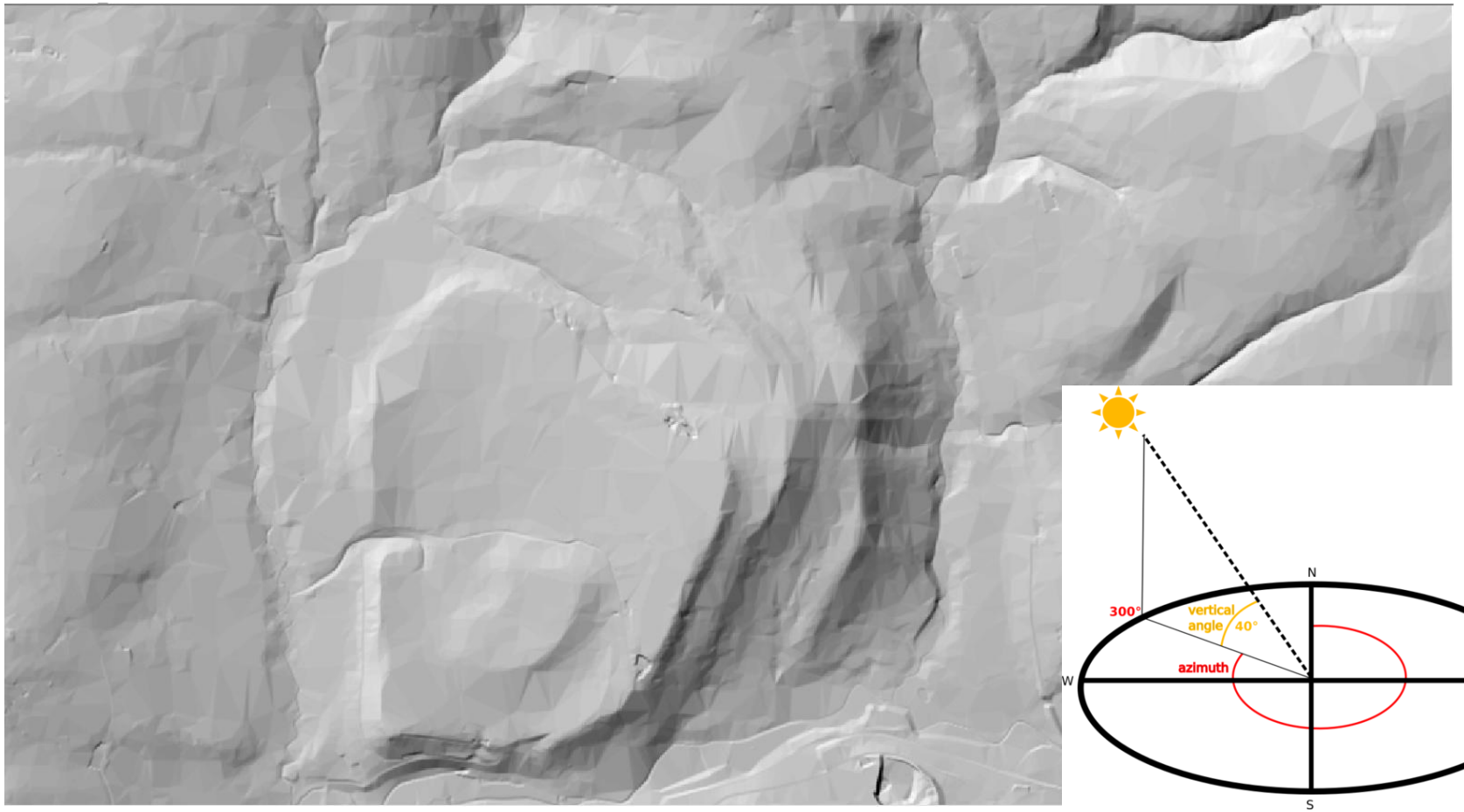
  Is a DEM augmented by other surface features including information about terrain shape (slope, curvature) and drainage structure.

DTM and DEM are often used interchangeably because they are similar

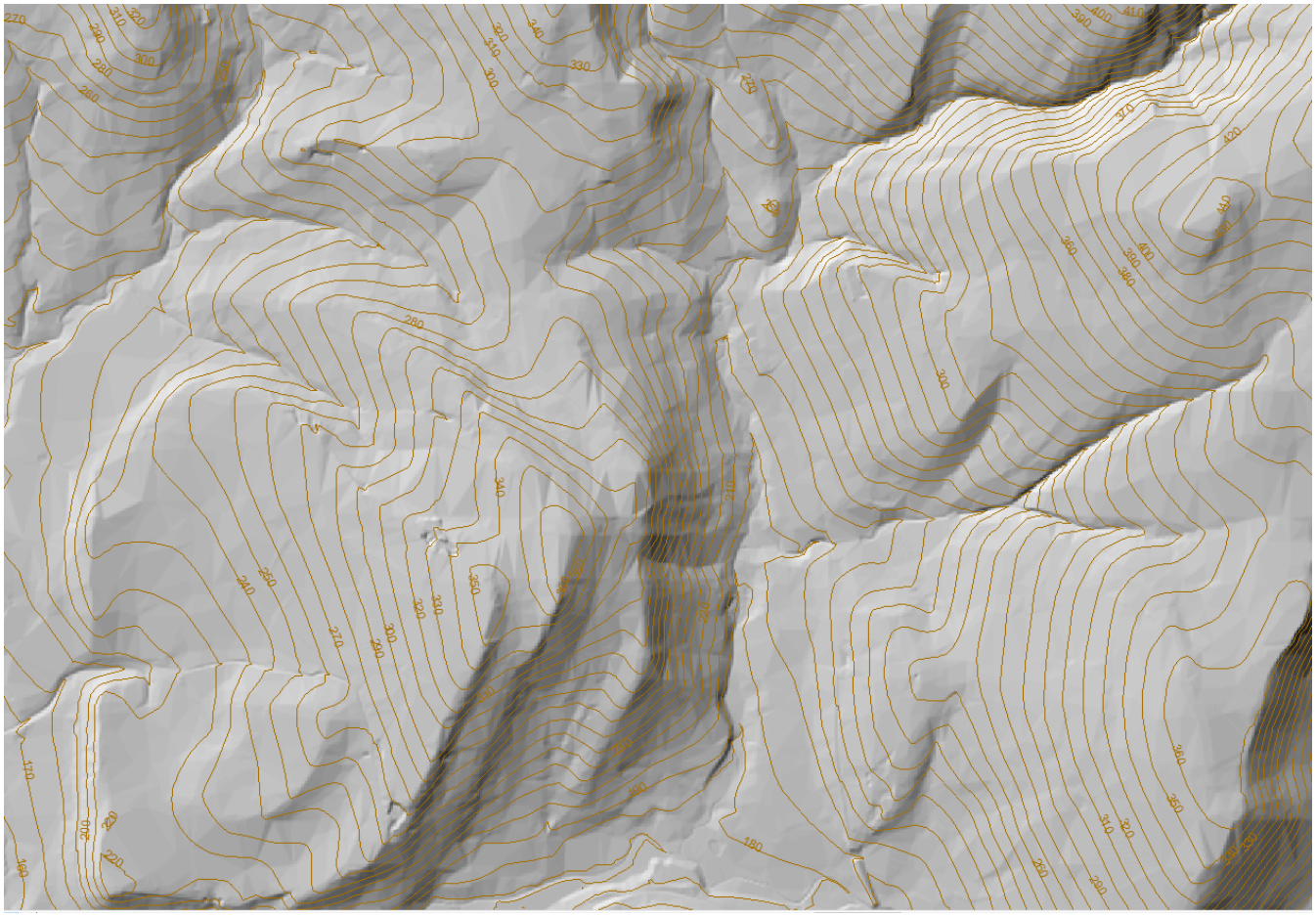
Digital Terrain Model



Topography (hillshade)



Relief and topography



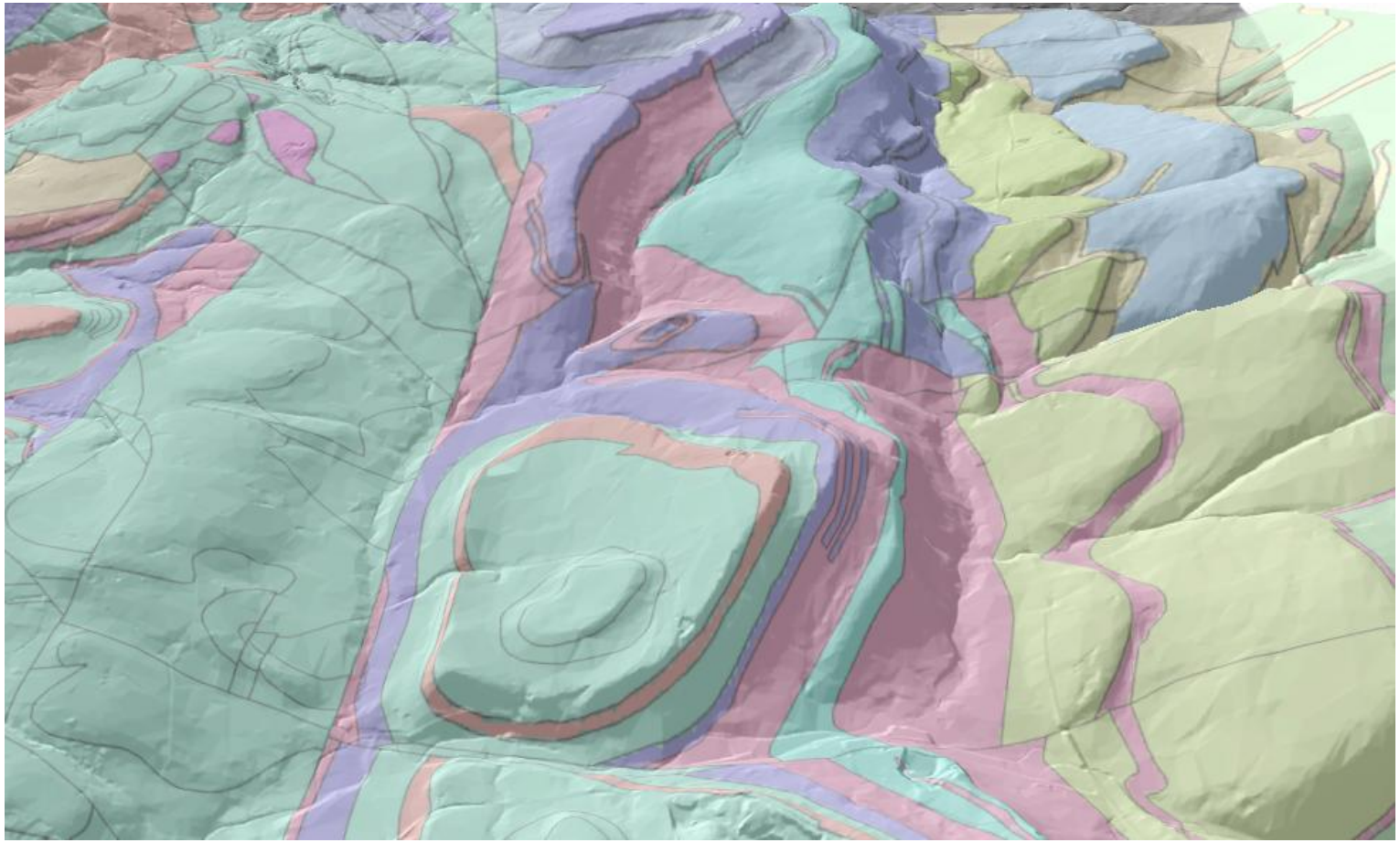
3D visualisation



Draping layers on a 3D surface

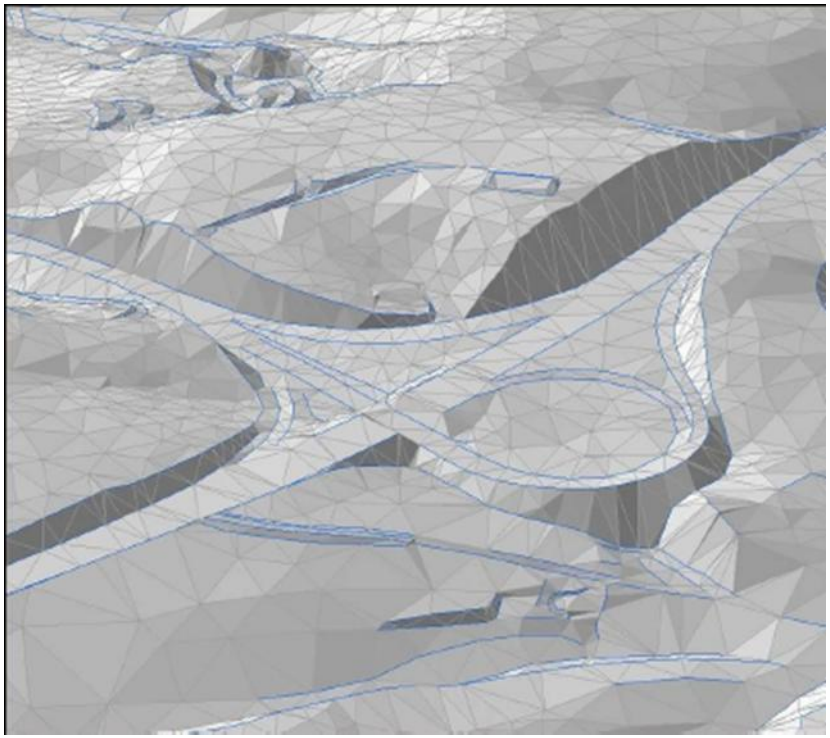


Draping layers on a 3D surface



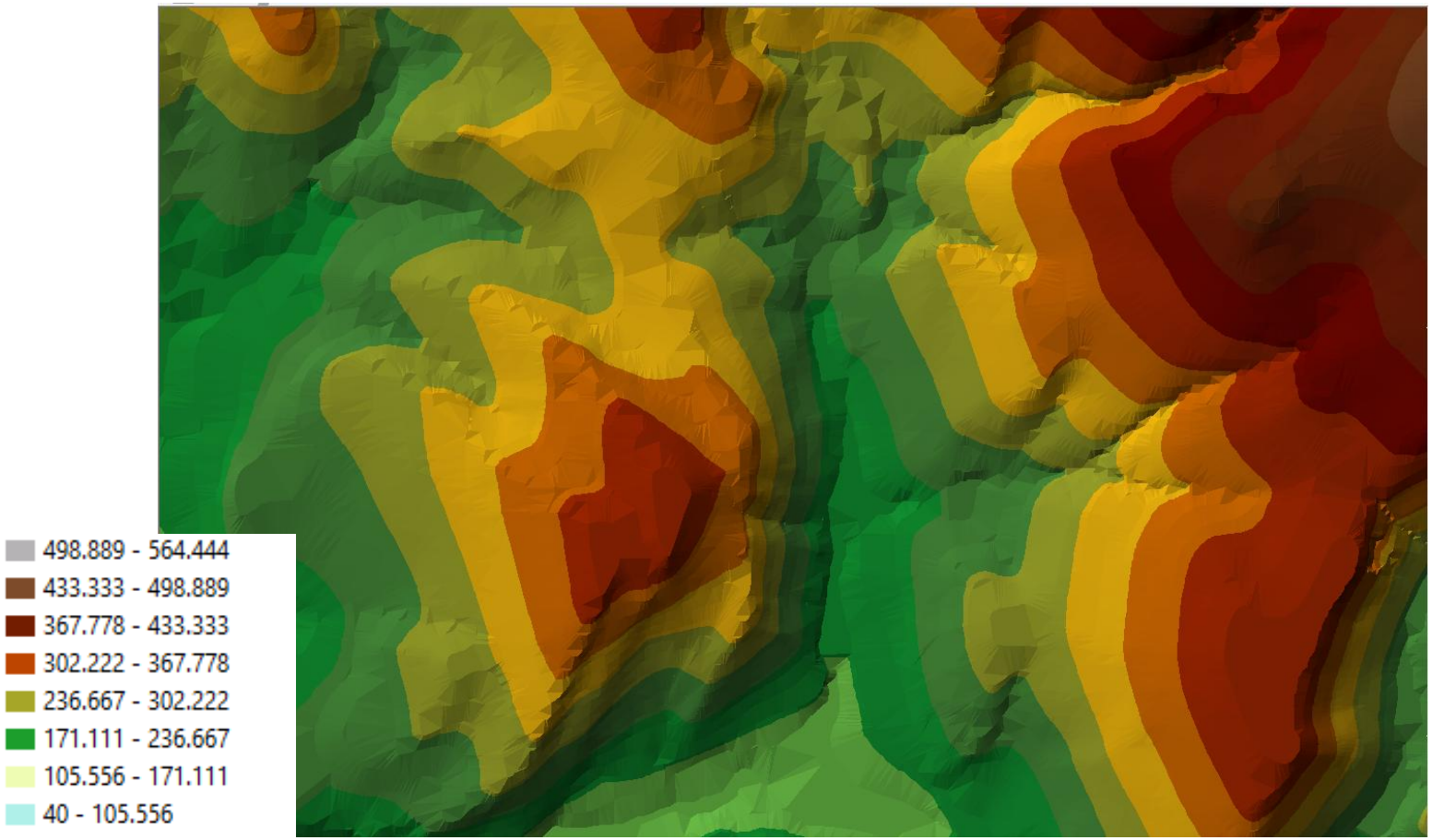
Triangulated irregular network (TIN)

- Vector-based model for surface morphology
- Surface is modelled as contiguous layer of triangles.
- Each triangle is a plane with constant gradient.
- Each point in TIN has 3D co-ordinates (x,y,z) as verticies (nodes).



OS Terrain 5m TIN – triangulated irregular networks uses triangles which relate to features more accurately than a grid. https://www.ordnancesurvey.co.uk/documents/os-terrain-5-user-guide.pdf

DEM using TIN



Map scale

The Ratio of distance on the map or image to the corresponding distance on the ground.

Smaller the map scale:

- the larger the ratio
- less accurate
- more abstract



Large and small map scales

- The smaller the map scale, the fewer geographical features and less detail about these geographical features can be captured and represented.

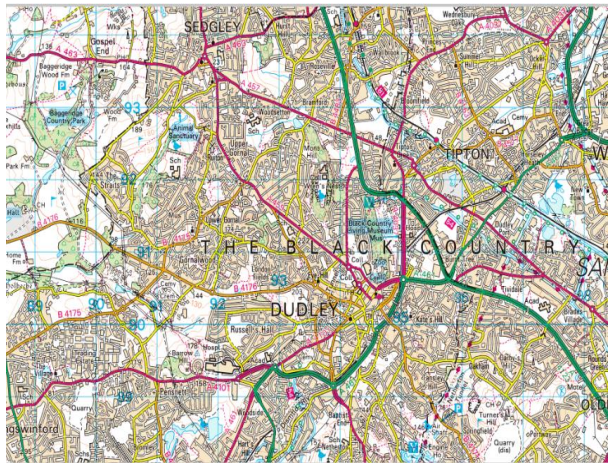


1:3,000,000

Small scale = large area Less detail



1:800,000

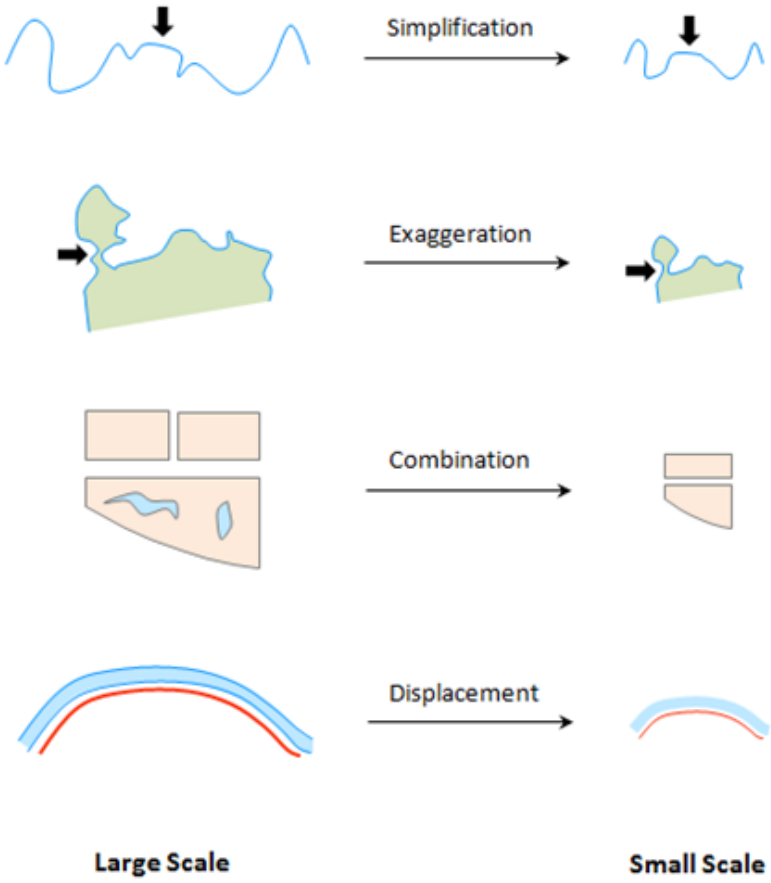


1:40,000

Large scale = small area More detail

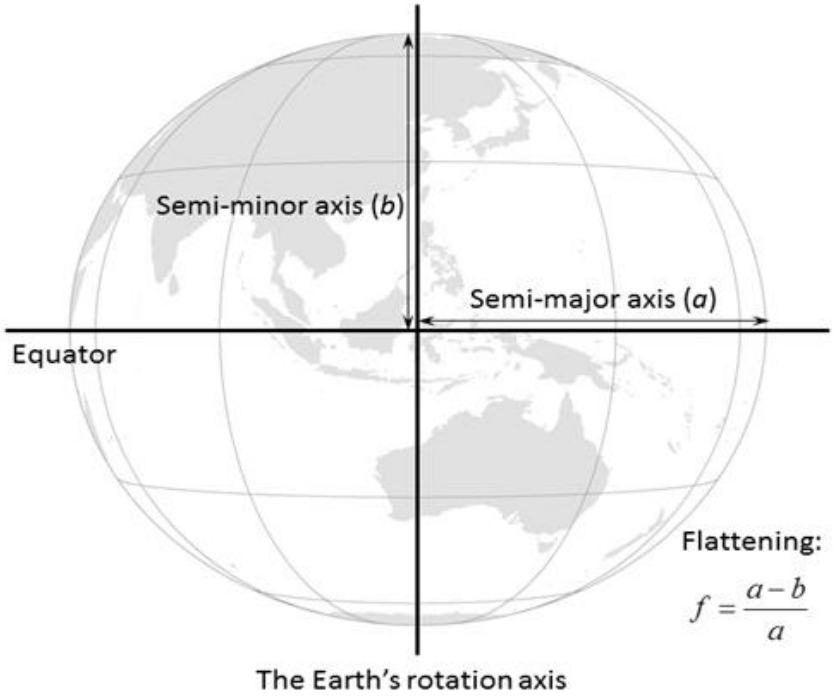
Generalisation

- Cartographical generalisation or abstraction is the process of transforming reality to a map.
- **Selection** chooses and retains the most important geographical features.
- **Simplification** means removing complexity
- **Exaggeration** is used to retain or emphasise important characteristics.
- **Combination** merges similar features.
- **Displacement** moves objects to keep them distinguishable.
- **Classification** group features by attributes



The shape of the Earth

geoid = an equipotential surface of the Earth's gravitational field= mean sea level
reference spheroid = smooth surface model of Earth's geoid



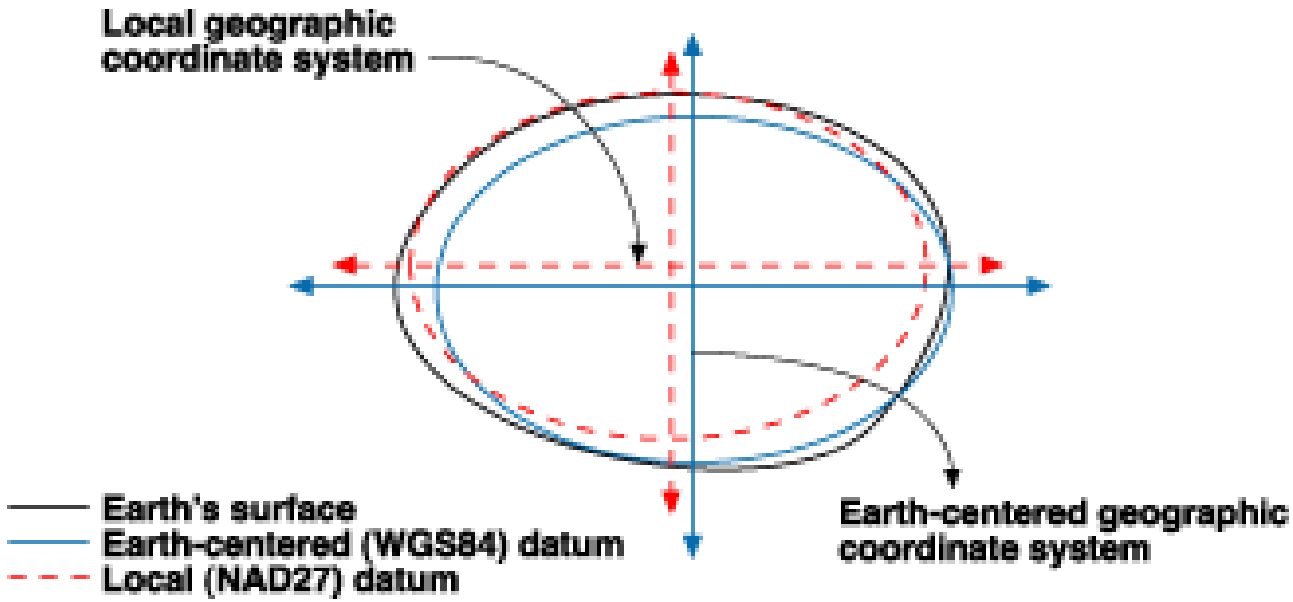
a - b = 21.4 km

Major Earth's spheroids

Name	Semi-major axis (m)	Semi-minor axis (m)	Flattening
Airy 1830	6,377,563.396	6,356,256.90923729	1/299.3249646
Bessel 1841	6,377,397.155	6,356,078.963	1/299.1528128
Clarke 1866	6,378,206.4	6,356,583.799999	1/294.9786982
Helmert 1906	6,378,200.0	6,356,818.16962789	1/298.3
International 1924	6,378,388.0	6,356,911.94612795	1/297.0
Krassowski 1940	6,378,245.0	6,356,863.01877305	1/298.3
Australian National	6,378,160.0	6,356,774.719	1/298.25
GRS 80	6,378,137.0	6,356,752.31414036	1/298.257222101
WGS 1984	6,378,137.0	6,356,752.3142	1/298.257223563

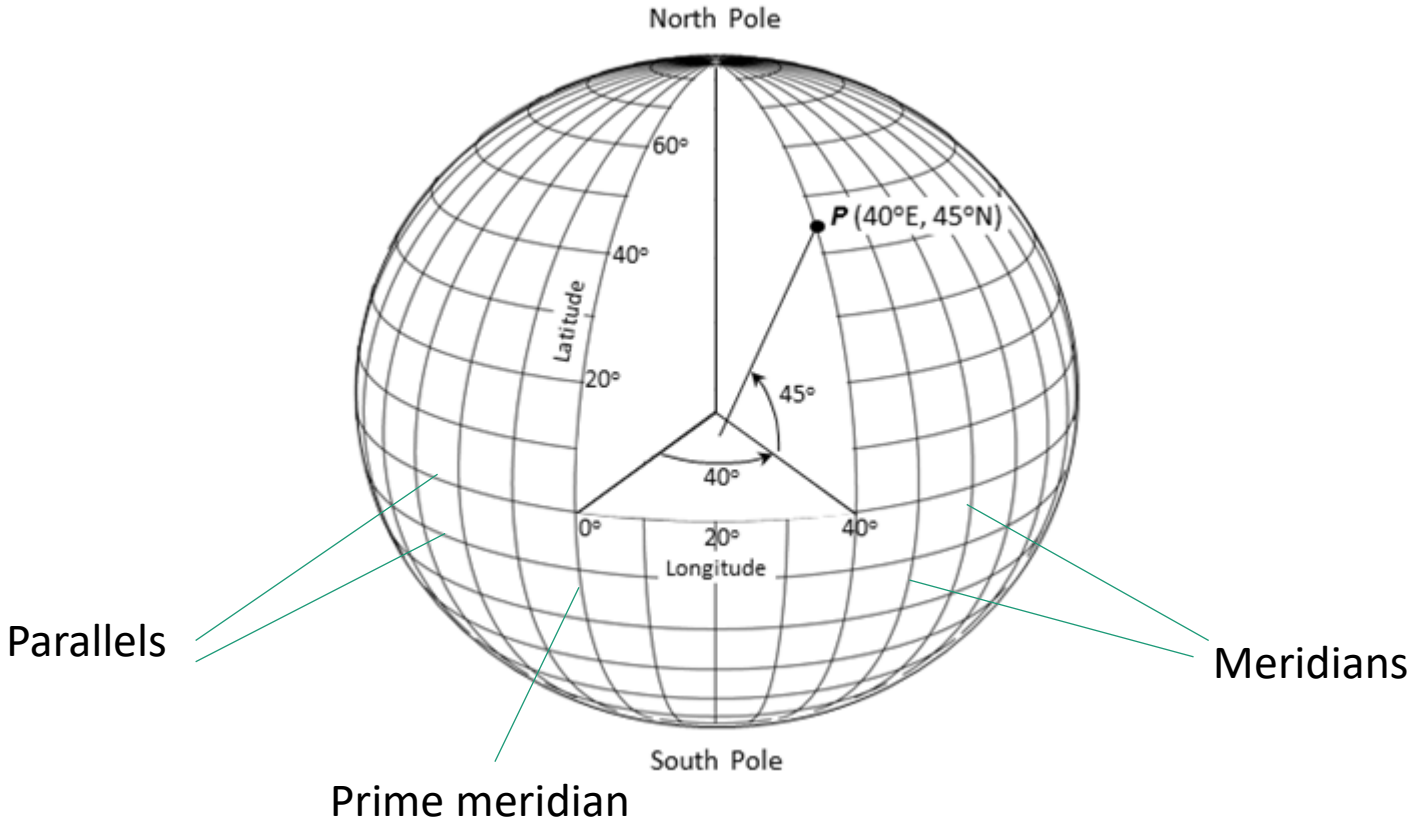
WGS = World Geodetic System
GRS = Geodetic Reference System (1980)

Datum accounts for local variations

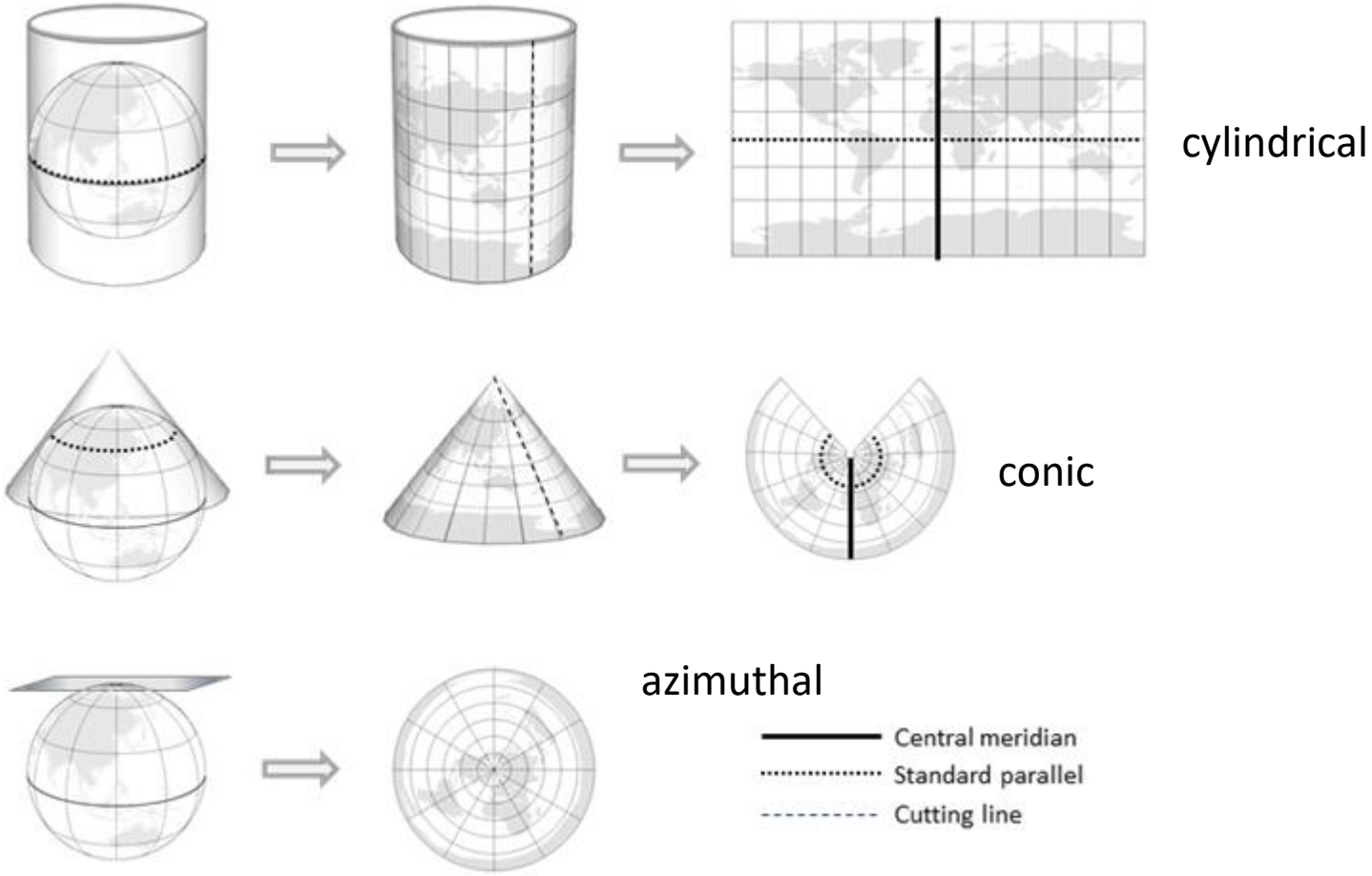


Geographical coordinate values depend upon the chosen reference spheroid and datum

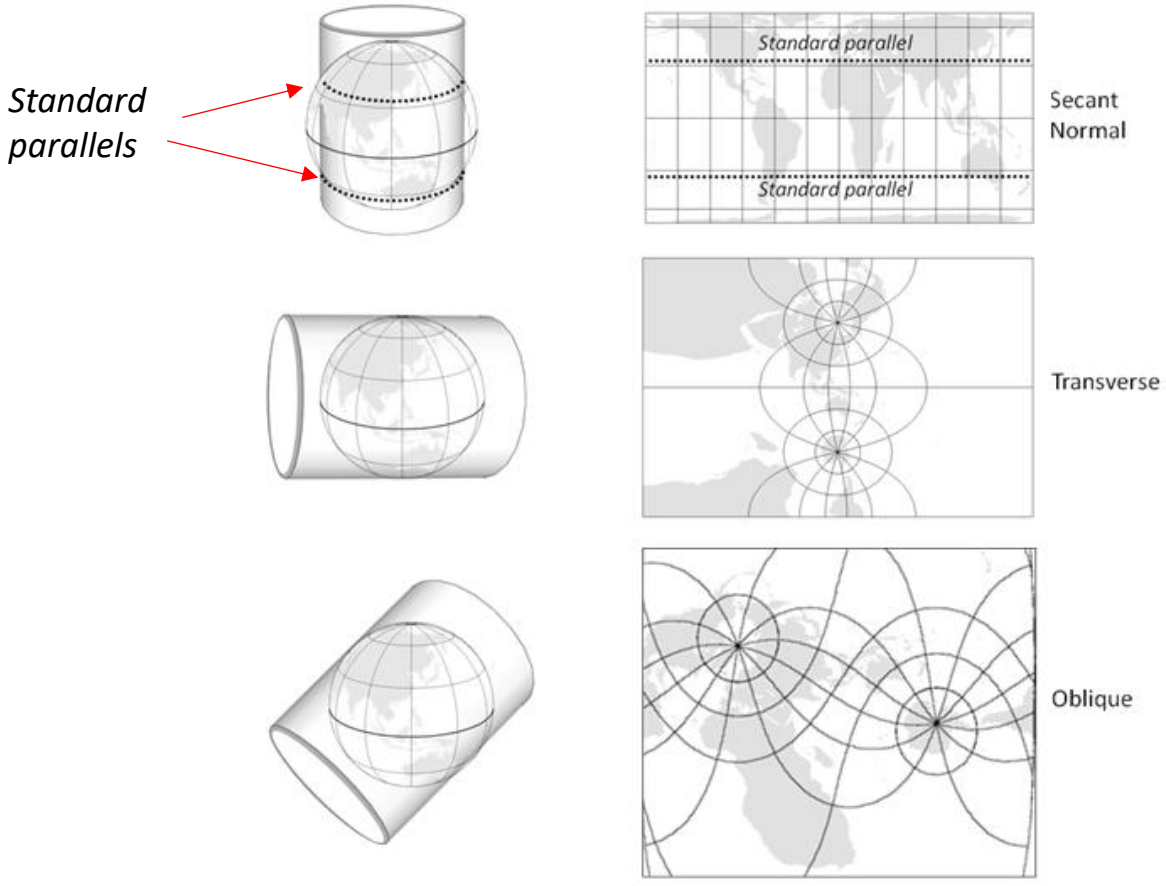
Geographic coordinate system



‘Normal’ map projections

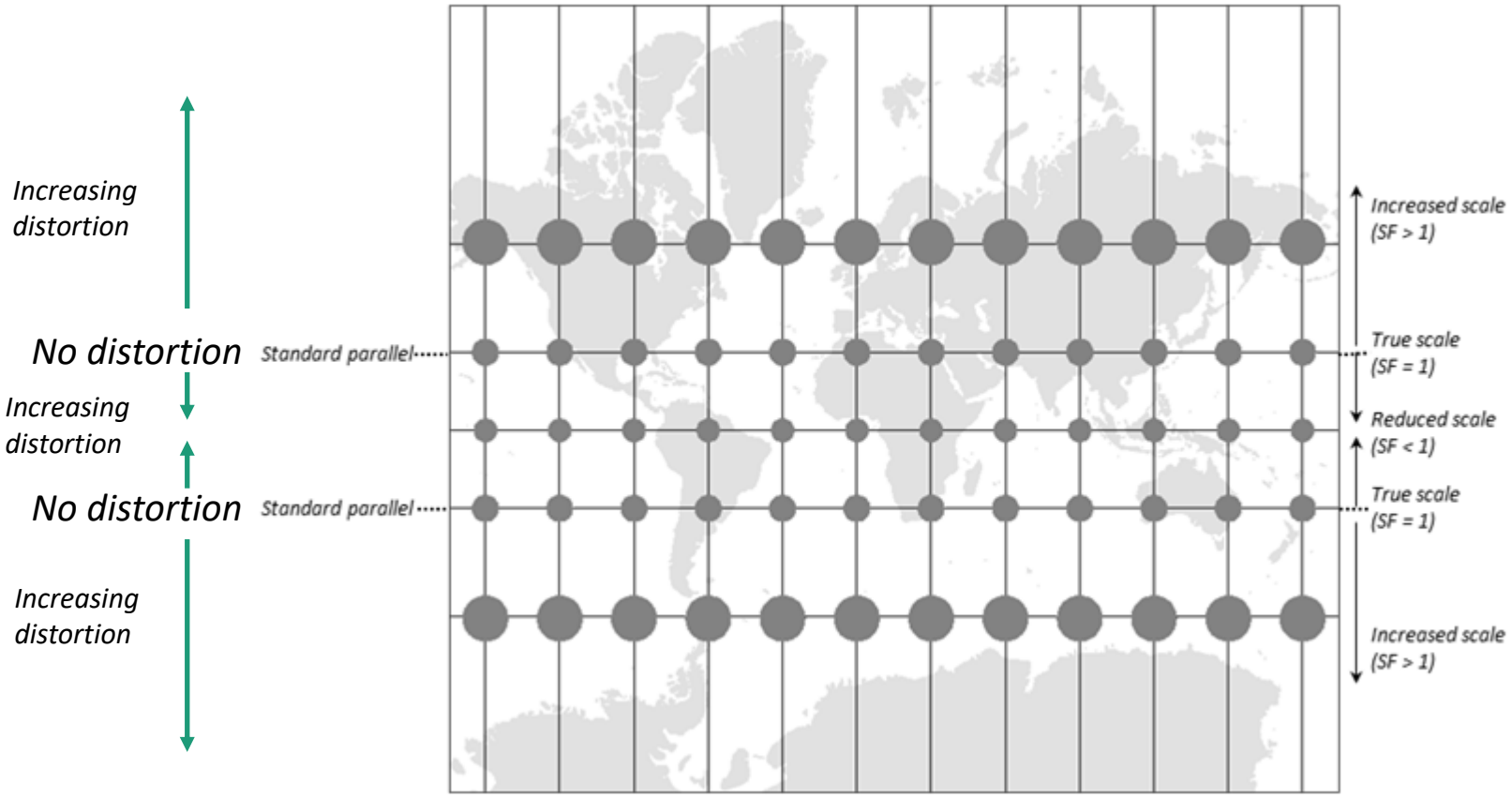


Cylindrical map projections



Secant lines cut the sphere

Secant Normal Mercator projection



Some countries are smaller than you think!

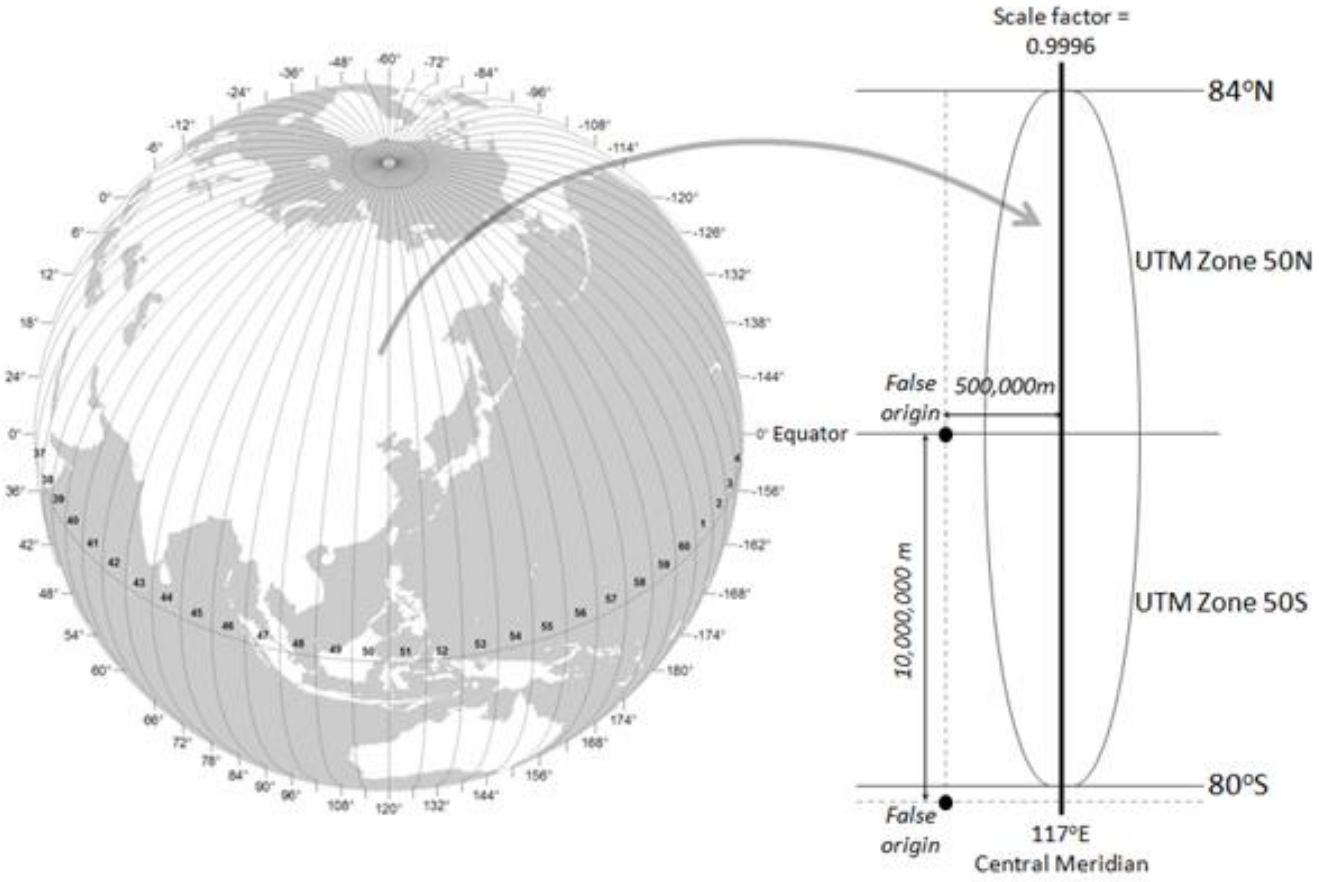


https://geospatialworld.net/blogs/maps-that-show-why-some-countries-are-not-as-big-as-they-look/

Mercator projection:

- preserves shapes and directions
- distorts area and distances (apart from equator)

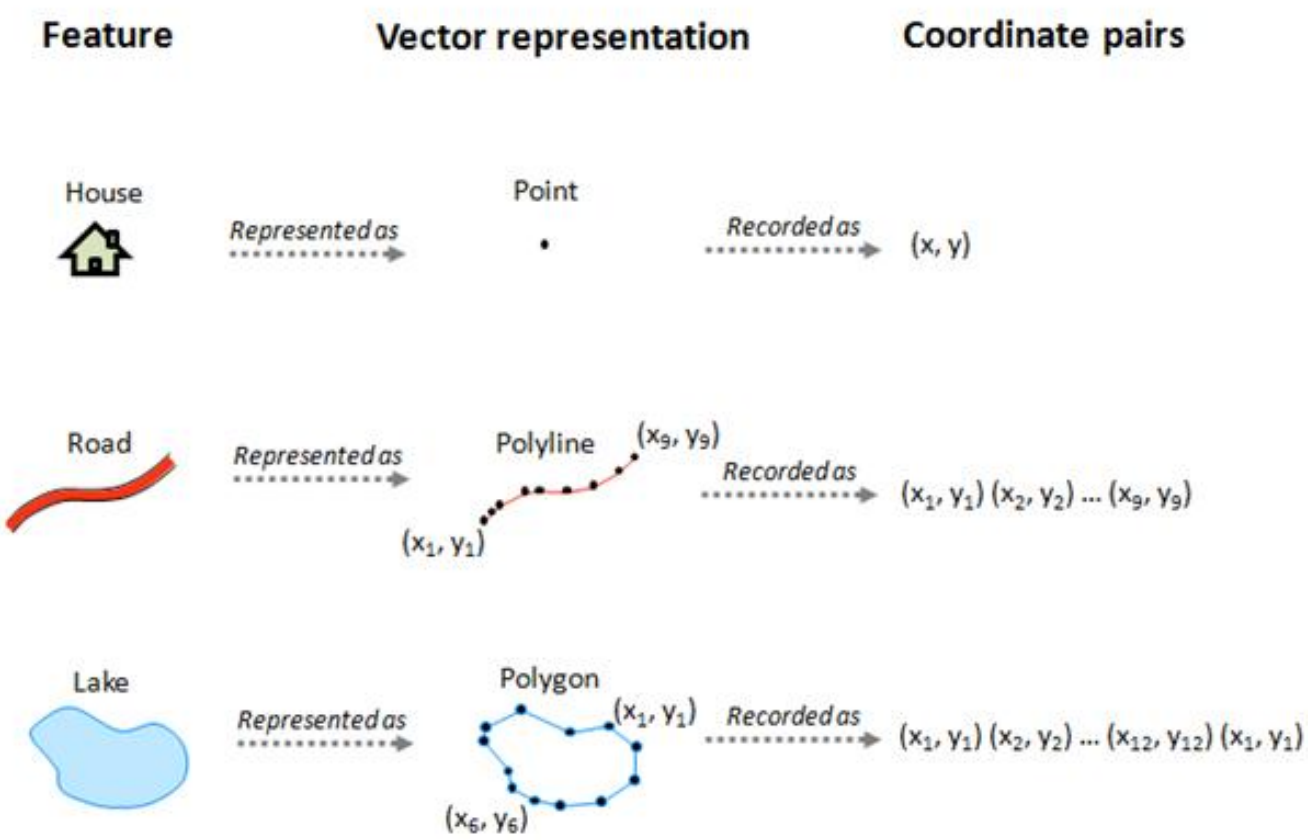
Universal Transverse Mercator (UTM) system



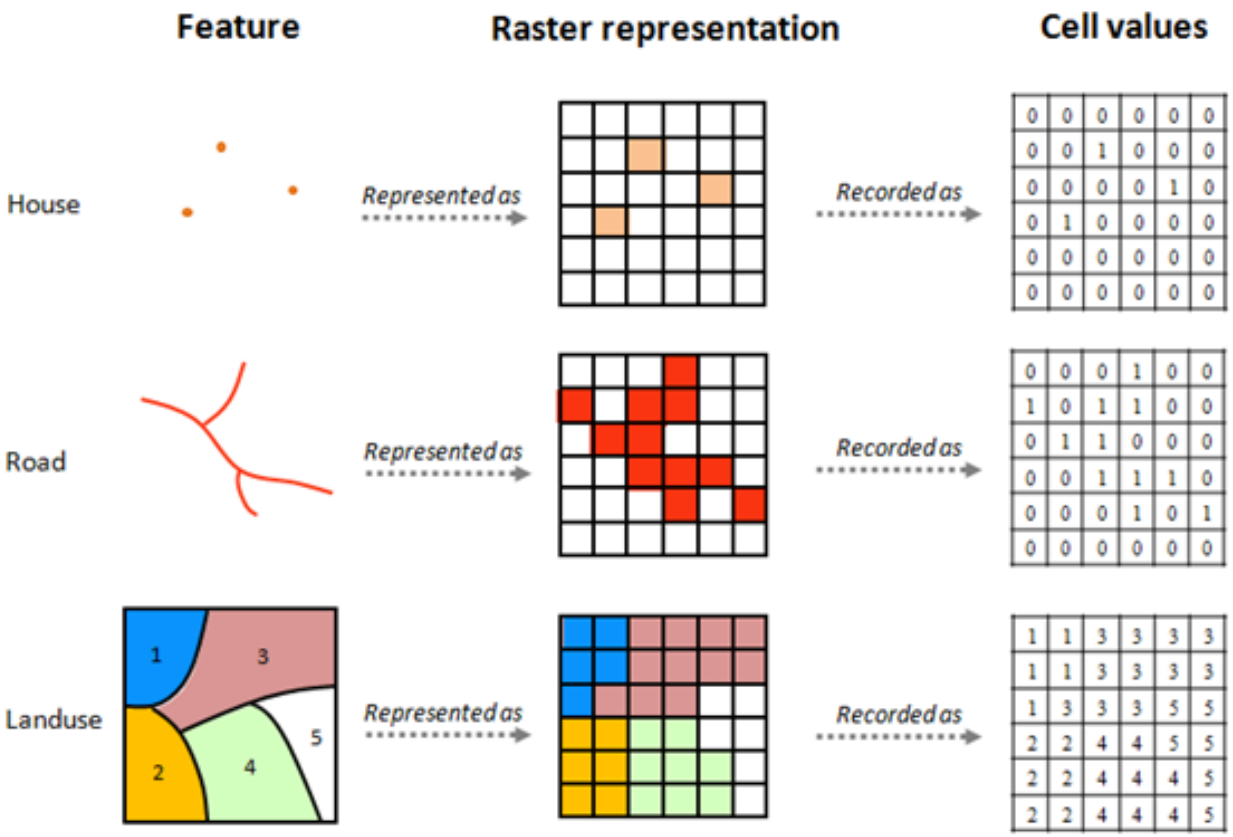
Vector data model is based on object view

- Based on the object view of geographical space.
- The vector data model typically stores spatial data as feature classes.
- A feature class is a collection of geographical features of the same geometric type (point, line and polygon) with the same set of attributes. Examples:
  - a **point** feature class may represent meteorological stations as points with observed values of meteorological variables;
  - a **line** feature class may represent streams with a set of stream attributes (length, elevation and gradient);
  - a **polygon** feature class may represent land parcels with land use types.

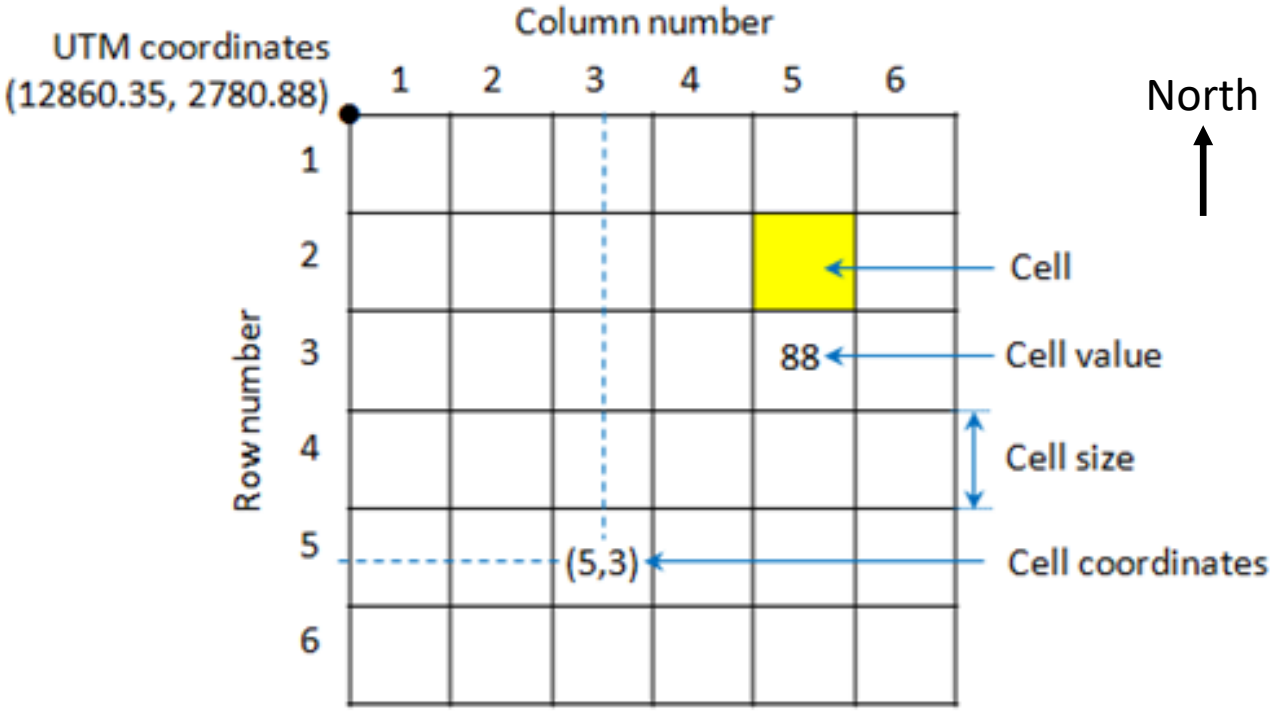
Vector data model coordinates



Raster data model is based on field view

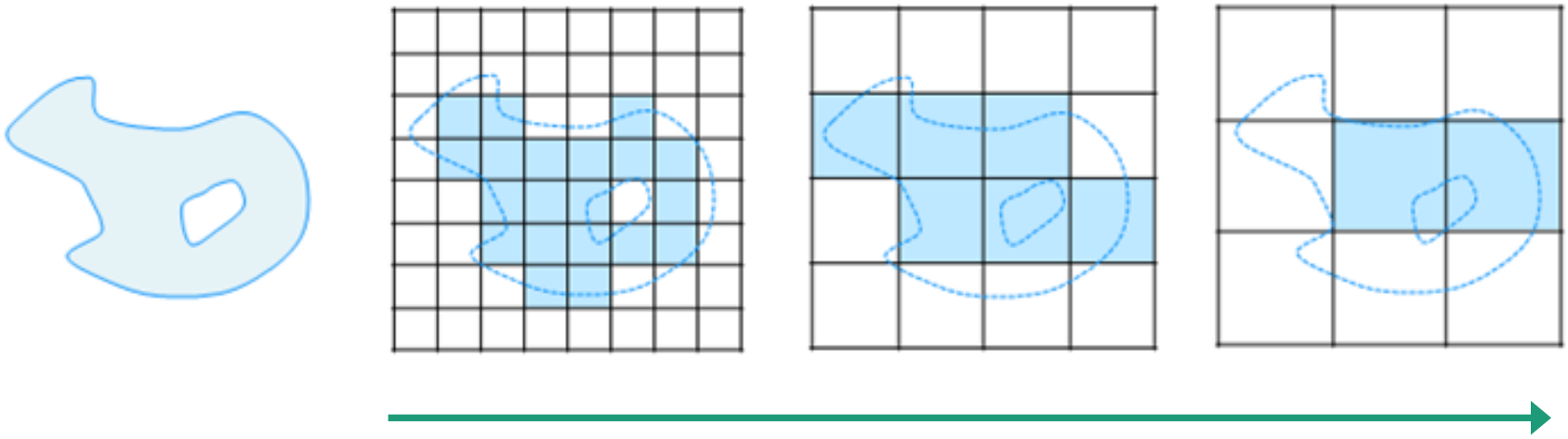


Elements of a raster



Cell or pixel depth:
greyscale: 8-bit = 2^8 -1 = 0-255 shades (black to white)
colour: 8-bit RGB channels = 16.7 million colours

Raster representations with different cell sizes



Increasing cell size = lower resolution and loss of detail

Homework and Resources

- Read Chapter 2 of course textbook: Spatial Data and Management
- Take the quiz to test your basic understanding (on Canvas)

Further resources

Introduction to Spatial Data | Esri Training Web Course

https://www.esri.com/training/catalog/5dd4485b6cd66f196fbeddad/introduction-to-spatial-data/

Read the text, don't attempt the ArcGIS Pro exercises